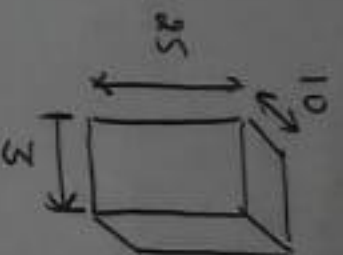


given... →

$$t = 3 \text{ m}$$

$$T_1 = 400^\circ \text{C}$$

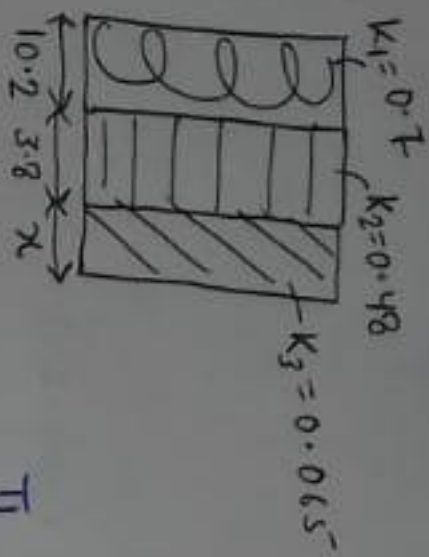
$$T_2 = 100^\circ \text{C}$$



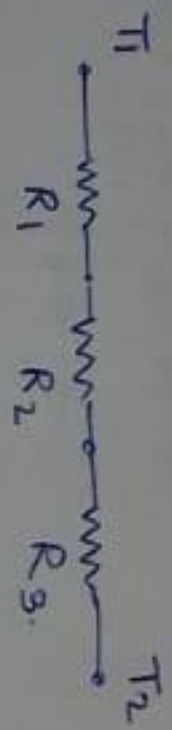
$$Q = -kA \frac{dT}{dx}$$

$$= -371.5 \times \frac{25 \times 10}{10000} \times \frac{300 \times 100}{3}$$

$$= 92 \text{ kW}$$



given -
With insulation = 0.2
Without insulation



$$Q_{\text{without insulation}} = \frac{T_1 - T_2}{R_1 + R_2} = \frac{T_1 - T_2}{\frac{10.2}{0.7 \times A} + \frac{3.8}{0.48 \times A}} \rightarrow \textcircled{1}$$

$$Q_{\text{with insulation}} = \frac{T_1 - T_2}{R_1 + R_2 + R_3} = \frac{T_1 - T_2}{\frac{10.2}{0.7 \times A} + \frac{3.8}{0.48 \times A} + \frac{x}{0.065 \times A}} \rightarrow \textcircled{2}$$

① ÷ ②

$$0.2 = \frac{(T_1 - T_2) \times \left(\frac{10.2}{0.7A} + \frac{3.8}{0.48A} \right)}{\left(\frac{10.2}{0.7 \times A} + \frac{3.8}{0.48 \times A} + \frac{x}{0.065 \times A} \right) \times (T_1 - T_2)}$$

$$0.2 = \frac{14.57 + 7.91}{14.57 + 7.91 + 15.38x}$$

$$3.076x = 22.48 - 4.496$$

$$x = 5.84 \text{ cm}$$

④

given $k_{insulation} = 0.17$

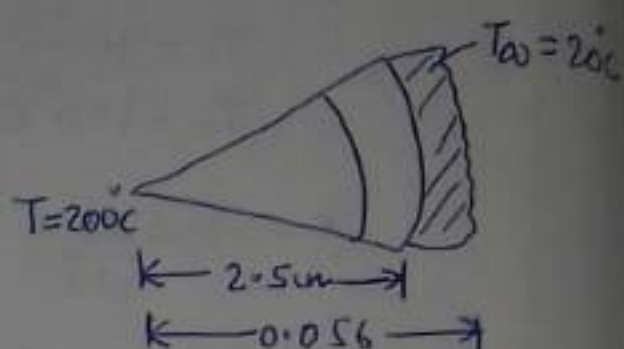
$$h = 3$$

$$r_c = \frac{k_{ins}}{h} = \frac{0.17}{3} = 0.056$$

$$Q_{\text{with insulation}} = \frac{(200 - 20) \times 2\pi L}{\frac{\ln\left(\frac{0.056}{0.025}\right)}{0.17} + \frac{1}{3 \times 0.056}}$$

$$= \frac{1130.97}{10.69}$$

$$= 105.79 \text{ W/m}$$



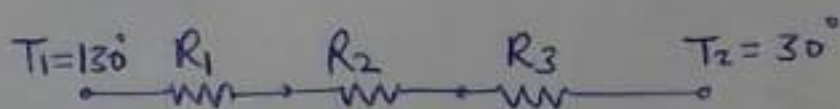
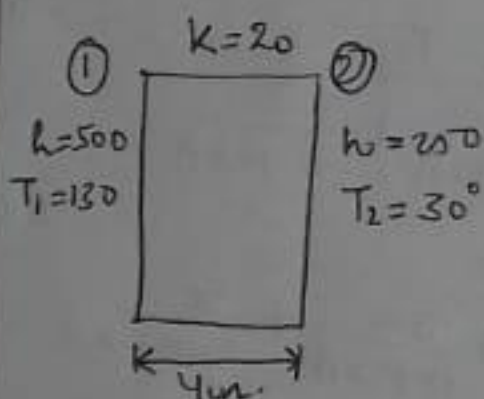
$$Q_{\text{without insulation}} = hA\Delta T$$

$$= 3 \times 2\pi \times \frac{2.5}{100} \times (200 - 20)$$

$$= 84.82$$

$$= 85 \text{ W.}$$

⑤



$$R_1 = \frac{1}{hA_s}$$

$$R_2 = \frac{L}{kA}$$

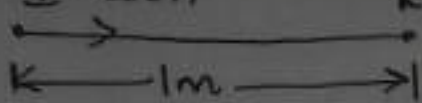
$$R_3 = \frac{1}{hA_s}$$

$$\frac{Q}{A} = \frac{130 - 30}{\frac{1}{500} + \frac{0.02}{20} + \frac{1}{250}}$$

$$= \frac{100}{2 \times 10^{-3} + 2 \times 10^{-3} + 4 \times 10^{-3}}$$

$$= 12.5 \text{ kW.}$$

(6)

$$I = 200 \text{ A} \quad k = 19$$


(3)

$$Q_g = Hg = I^2 R$$

$$= (200)^2 \times \frac{70 \times 10^{-6} \times 1}{\frac{\pi}{4} \times \left(\frac{3}{1000}\right)^2}$$

$$= 3961 \text{ J/s.}$$

$$Q = hA(T_w - 110)$$

$$3961 = 4 \times 10^3 \times 21\pi \times \frac{1.5}{1000} \times 1 \times (T_w - 110)$$

$$T_w = 215.06^\circ \text{C.}$$

$$T_o = T_w + q_g R^2 / 4k.$$

$$= 215.06 + \frac{3961 \times \left(\frac{1.5}{1000}\right)^2}{\pi \times \left(\frac{1.5}{1000}\right)^2 \times 1} \bigg/ 4 \times 19$$

$$= 231.06^\circ \text{C.}$$

$$T_T = 170^\circ \text{C}$$

$$T_a = 25^\circ \text{C.}$$

using data book:-

from graph:- $L \sqrt{2h/k} = \frac{1.5}{100} \sqrt{\frac{2 \times 130}{200 \times 1 \times 10^{-3}}}$

$$= 0.54$$

$$\frac{r_o}{r_i} = 2.2$$

from efficiency graph-

$$\text{eff } \eta = 85\%$$

$$Q_{act} = \eta \cdot Q_{max}$$

$$= 0.85 \times hA\Delta T$$

$$= 0.85 \times 130 \times \pi (r_2^2 - r_1^2) \times 2 \times (170 - 25)$$

$$= 0.85 \times 130 \times \pi \times ((1.5 + 1.25)^2 - (1.25)^2) \times 2 \times (170 - 25)$$

$$= 60 \text{ W.}$$

9.

$$h = 10$$

$$t = ?$$

(4)

$$Bi = \frac{hL}{k} = \frac{10 \times 4\pi \times (0.025)^2}{\frac{4}{3} \times \pi \times (0.025)^3 \times 35}$$

$$= < 0.01$$

$$\therefore \frac{T - T_{\infty}}{T_0 - T_{\infty}} = e^{\frac{-hA_s t}{\rho V C}}$$

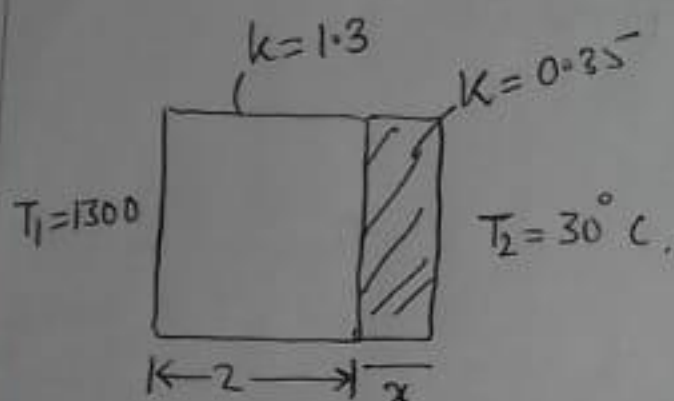
$$\frac{150 - 100}{450 - 100} = e^{\frac{-3 \times 10 \times 4\pi \times (0.025)^2 \times t}{7753 \times 4\pi \times (0.025)^3 \times 0.46}}$$

$$0.142 = e^{-1.40t}$$

$$1.951 = 1.40t$$

$$t = 1.39 \text{ sec.}$$

10.



given -

$$\frac{Q}{A} = 1830 \text{ W/m}^2.$$

$$\therefore Q = \frac{T_1 - T_2}{R_1 + R_2}$$

$$\frac{Q}{A} = \frac{1300 - 30}{\frac{L}{k} + \frac{x}{K}} = 1830$$

$$1830 = \frac{1300 - 30}{\frac{0.02}{1.3} + \frac{x}{0.35}}$$

$$1830 = \frac{1300 - 30}{0.0153 + 2.85x}$$

$$27.99 + 5215.5x = 1270$$

$$x = 0.238 \text{ m}$$

$$= 23.8 \text{ cm.}$$

given $\rightarrow$

(5)

$$K = K_0 (1 + \beta T^2)$$

$$Q = -KA \frac{dT}{dx}$$

$$\int_0^x Q \cdot dx = \int_{T_1}^T -KA dT$$

$$Q \cdot x = - \int_{T_1}^T K_0 (1 + \beta T^2) A dT$$

$$Qx = -K_0 A \int_{T_1}^T dT + \beta T^2 dT$$

$$Qx = -K_0 A \left[(T - T_1) + \left[\beta \frac{T^3}{3} \right]_{T_1}^T \right]$$

$$Qx = -K_0 A \left[T - T_1 + \beta \frac{T^3}{3} - \beta \frac{T_1^3}{3} \right]$$

$$Qx = -\frac{K_0 A}{3} [3T - 3T_1 + \beta(T^3 - T_1^3)] \rightarrow (1)$$

also $x=L, T=T_2$

$$QxL = -\frac{K_0 A}{3} [3T_2 - 3T_1 + \beta(T_2^3 - T_1^3)] \rightarrow (2)$$

 $\therefore Q$ is same $\rightarrow$

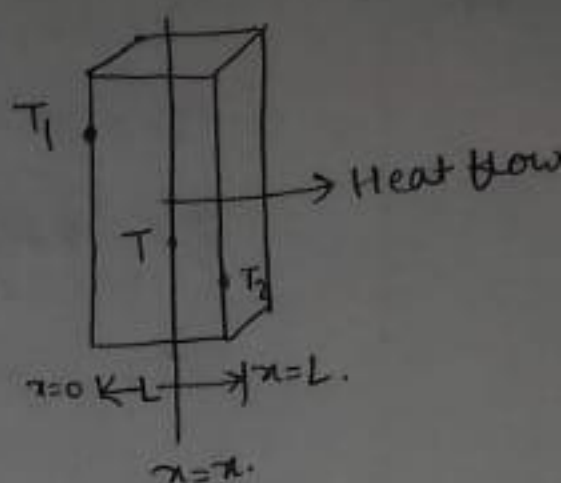
also eq. (2)

$$Q = -\frac{K_0 A}{3L} [3T_2 - 3T_1 + \beta(T_2^3 - T_1^3)]$$

$$Q = \frac{3T_2 - 3T_1 + \beta(T_2^3 - T_1^3) - K_0 A}{3L}$$

$$Q = \frac{(T_2 - T_1)}{L} (3 + \beta(T_2^2 + T_1^2 + T_1 T_2)) - K_0 A$$

$$Q = \frac{\left(\frac{T_1 - T_2}{L} \right)}{\left[3 + \beta(T_2^2 + T_1^2 + T_1 T_2) \right] K_0 A} \quad \underline{\underline{Q}}$$



14.

$$k = 0.04 \text{ W/mK}$$

$$T = 150x^2 - 30x$$

$$\therefore Q = -kA \frac{dT}{dx}$$

$$Q = -0.04 \times A \left| \frac{d}{dx} (150x^2 - 30x) \right|_{x=0}$$

$$\frac{Q}{A} = -0.04 \times (150 \times 2x - 30)$$

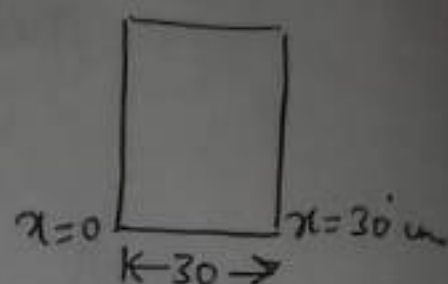
$$\text{at } x=0$$

$$Q = 1.2 \text{ W}$$

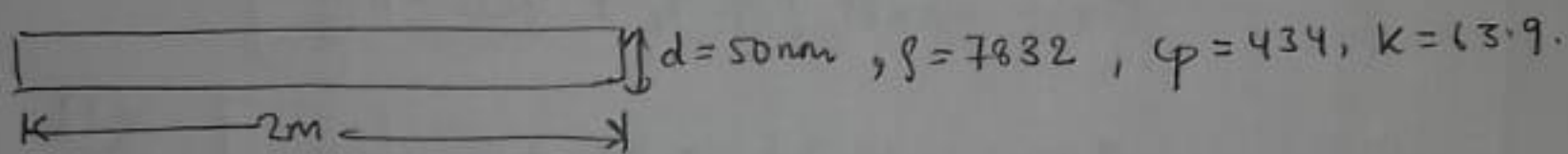
$$\text{at } x=30$$

$$Q = -0.04 \times (150 \times 2 \times 30 - 30)$$

$$Q = -358.8 \text{ W} \underline{\underline{A}}$$



15.



$$T_1 = 850^\circ\text{C}$$

$$T_2 = 95^\circ\text{C}$$

$$T_\infty = 40^\circ\text{C}$$

$$h = 450$$

$$Bi = \frac{hL}{k} = \frac{450 \times \pi \times \left(\frac{25}{1000}\right)^2 \times 2}{63.9 \times 2 \times \pi \times \left(\frac{25}{1000}\right)^2}$$

$$= 0.08 \underline{\underline{a}}$$

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-hAt/\rho V c}$$

$$\frac{95 - 40}{850 - 40} = e^{\frac{-450 \times 2 \times \pi \times 0.025^2 \times 2 \times t}{7832 \times \pi \times 0.025^2 \times 2 \times 434}}$$

$$0.0679 = e^{-0.01059 t}$$

$$t = 251 \text{ sec.}$$

16.

$$T_i = 20^\circ\text{C}$$

$$T_a = -15^\circ\text{C}$$

Let depth $x \rightarrow$

$$\frac{T - T_a}{T_i - T_a} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right]$$

$$\frac{0 + 15}{20 + 15} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right] = 0.4285$$

$$\frac{x}{2\sqrt{\alpha t}} \approx 0.44 \quad (\text{from data book})$$

$$x = 0.44 \times 2\sqrt{0.138 \times 10^{-6} \times 60 \times 24 \times 60 \times 60}$$

$$x = 0.60 \text{ m.}$$

17.

$$T_i = 35^\circ\text{C}$$

$$T_a = 250^\circ\text{C}$$

$$x = 2.5 \text{ m}$$

$$t = 0.5 \text{ min} = 0.5 \times 60 = 30 \text{ sec.}$$

$$\frac{T - T_a}{T_i - T_a} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right]$$

$$\frac{T - 250}{35 - 250} = \text{erf} \left[\frac{2.5}{2\sqrt{\alpha t}} \right] = \text{erf}(0.609)$$

$$\frac{T - 250}{35 - 250} = 0.6059 \quad (\text{from data book})$$

$$T = 118.5^\circ\text{C} \underline{\underline{=}}$$

18.

(8)

$$x = 0.2m$$

$$d = 15cm$$

$$l = 4m$$

$$K = 0.8$$

$$T = 5^\circ C.$$

$$\begin{aligned} \therefore S &= \frac{2\pi L}{\cosh^{-1}(D/r)} \\ &= \frac{2\pi \times 4}{\cosh^{-1}(20/7.5)} = 15.35m \end{aligned}$$

$$\begin{aligned} Q &= K S \Delta T = 0.8 \times 15.38 \times (75-5) \\ &= 859.6W. \end{aligned}$$

19.

$$\therefore \text{given - } 50 \times 50 \times 50 \text{ cm.}$$

$$T = 500$$

$$T_2 = 50^\circ$$

$$\text{thickness} = 10$$

$$Q = -K A \frac{dT}{dx}$$

$$S_{\text{wall}} = \frac{A}{L} = \frac{0.5 \times 0.5}{0.1} = 2.5m$$

$$S_{\text{edge}} = 0.54 D = 0.54 \times 0.5 = 0.27m$$

$$S_{\text{corner}} = 0.15 \times L = 0.15 \times 0.1 = 0.015m$$

$$A = \text{Area of wall}$$

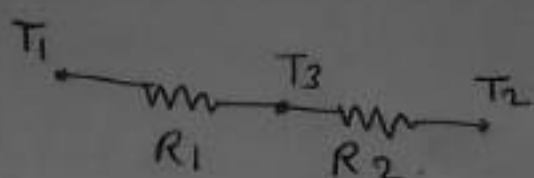
$$L = \text{thickness}$$

$$D = \text{length edge}$$

$$\begin{aligned} S &= 6 \text{ wall section} \times 2.5 + 12 \text{ edge} \times 0.27 + 8 \text{ corner} \times 0.015 \\ &= 18.36 \end{aligned}$$

$$\begin{aligned} Q &= K S \Delta T = 1.04 \times 18.36 \times (500 - 50) \\ &= 8.59 \text{ kW} \end{aligned}$$

⑨



$$R_1 = \frac{\ln(r_2/r_1)}{2\pi k L}$$

$$R_2 = \frac{\ln(r_3/r_2)}{2\pi k L}$$

$$\frac{Q}{L} = \frac{(T_1 - T_2) 2\pi}{\frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2}}$$

$$\frac{Q}{L} = \frac{500 \times 2\pi}{\frac{\ln(2/1)}{19} + \frac{\ln(5/2)}{0.2}}$$

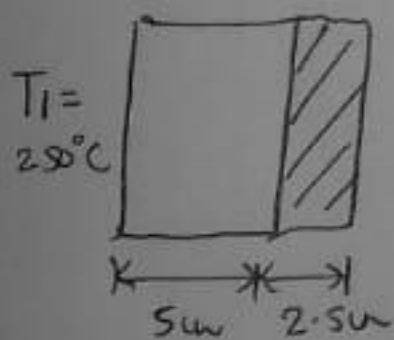
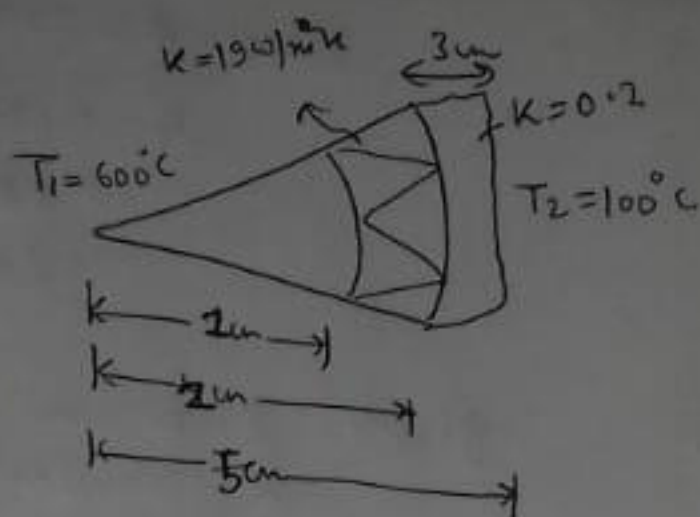
$$\frac{Q}{L} = 680 \text{ W/m } \underline{a}$$

Also,

$$Q = \frac{T_1 - T_3}{R_1}$$

$$680 = \frac{(600 - T_3) \times 2\pi \times 19}{\ln 2}$$

$$T_3 = 596.05^\circ\text{C } \underline{a}$$



$Q = 52 \text{ kW}$ given
area = ?



$$Q = 52 \times 10^3 = \frac{250 - 35}{\frac{0.05}{406} + \frac{0.025}{0.04}}$$

$$= \frac{215}{1.23 \times 10^{-4} + 0.625} = \frac{343.93}{\text{A}}$$

$$52 \times 10^3 A = 343.93, \quad (10)$$

$$A = 6.614 \times 10^{-3} \text{ m}^2$$

$$= 66.14 \text{ cm}^2$$

$$= 66.14 \text{ cm}^2 \underline{\underline{Ans}}$$

Q7:

given, $L = 7.5 \text{ cm}$

$k = 200 \text{ W/m}^\circ\text{C}$

thickness = 3 mm

$$L_c = L + t/2 = 7.5 + 0.15 = 7.65 \text{ cm}$$

$$m = \sqrt{\frac{hP}{kA}} = \left[\frac{h(2z + 2t)}{k + z} \right]^{1/2} \approx \sqrt{\frac{2h}{kt}}$$

$$m = \left[\frac{2 \times 10}{200 \times 3 \times 10^{-3}} \right]^{1/2} = 5.774$$

also,

$$q = (\tanh mL_c) \sqrt{hPkA} \theta_0$$

$$A = 1 \times 3 \times 10^{-3} = 3 \times 10^{-3} \text{ m}^2$$

(depth = 1 m).

$$q = 5.774 \times 200 \times 3 \times 10^{-3} \times (200 - 50) \tanh [5.774 \times 0.0765]$$

$$= 355 \text{ W/m} \underline{\underline{Ans}}$$